20 RATES OF REACTION

Reactions can vary in speed between those that happen within fractions of a second – explosions, for example – and those that never happen at all. Gold can be exposed to the air for thousands of years and not react in any way.

This chapter looks at the factors controlling the speeds of chemical reactions.



▲ Figure 20.1 Some reactions are very fast.



▲ Figure 20.2 Some reactions happen over several minutes.



▲ Figure 20.3 Rusting takes days or weeks.



▲ Figure 20.4 The weathering of limestone and the formation of stalagmites and stalactites takes a very long time.

LEARNING OBJECTIVES

- Describe experiments to investigate the effects of changes in the surface area of a solid, the concentration of a solution, the temperature and the use of a catalyst on the rate of a reaction
- Describe the effects of changes in the surface area of a solid, the concentration of a solution, the pressure of a gas, the temperature and the use of a catalyst on the rate of a reaction
- Explain the effects of changes in the surface area of a solid, the concentration of a solution, the pressure of a gas and the temperature on the rate of a reaction in terms of particle collision theory
- Know that a catalyst is a substance that increases the rate of a reaction, but is chemically unchanged at the end of the reaction
- Know that a catalyst works by providing an alternative pathway with lower activation energy

CHEMISTRY ONLY

- Draw and explain reaction profile diagrams showing ΔH and activation energy.

- Practical: Investigate the effect of changing the surface area of marble chips and of changing the concentration of hydrochloric acid on the rate of reaction between marble chips and dilute hydrochloric acid
- Practical: Investigate the effect of different solids on the catalytic decomposition of hydrogen peroxide solution

EXPERIMENTS TO MEASURE THE RATE OF REACTION

The **rate** of a reaction is the speed at which the amount of reactants decreases or the amount of products increases. It is measured as *a change in the concentration (or amount) of reactants or products per unit time* (per second, per minute etc.).

$$\text{rate of reaction} = \frac{\text{change in concentration, volume or mass}}{\text{time}}$$

Safety Note: Wear eye protection and avoid skin contact with the acid.

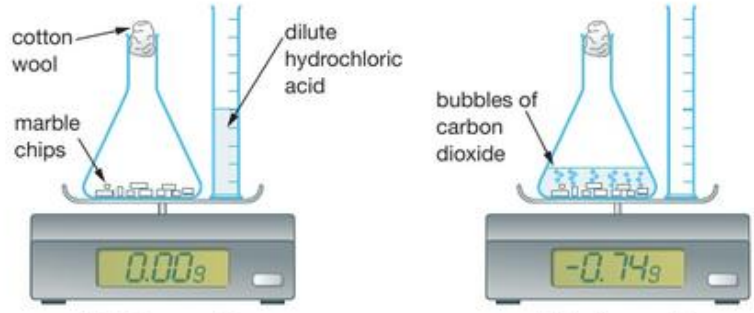
ACTIVITY 5

▼ PRACTICAL: AN INVESTIGATION OF THE RATE OF REACTION BETWEEN MARBLE CHIPS AND DILUTE HYDROCHLORIC ACID

Marble chips are made of calcium carbonate and react with hydrochloric acid to produce carbon dioxide gas. Calcium chloride solution is also formed.

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

Figure 20.5 shows some apparatus that can be used to measure how the mass of carbon dioxide produced changes with time. Part (a) is drawn as the apparatus would look before the reaction starts.



▲ Figure 20.5 Investigating the reaction between calcium carbonate and hydrochloric acid.

The following procedure could be used:

- Use a measuring cylinder to measure 25 cm^3 of 2.00 mol/dm^3 dilute hydrochloric acid.
- Add 5.00 g of large marble chips to a conical flask and place a piece of cotton wool at the opening of the flask. The cotton wool is there to allow the carbon dioxide to escape during the reaction, but to stop any acid spitting out. The marble is in *excess* – some of it will be left over when the acid is all used up.
- Place everything on a balance and reset it to zero.
- Add the acid to the marble chips and record the reading on the balance every 30 seconds.

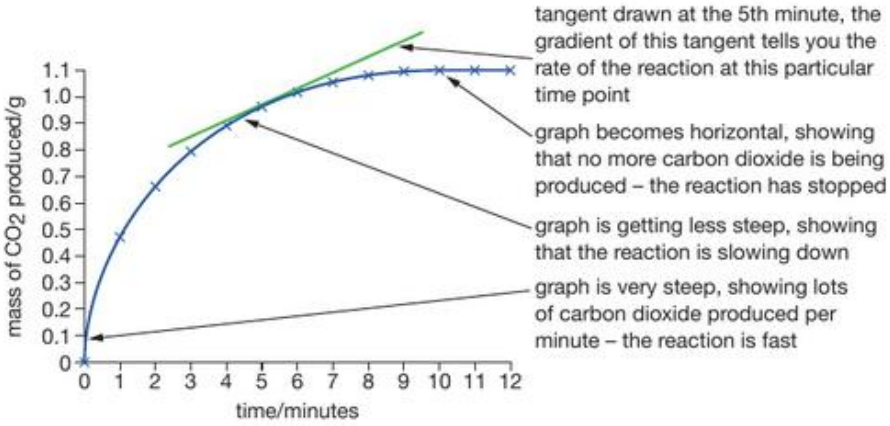
Part (b) shows what happens during the reaction. The acid has been poured into the flask and everything has been replaced on the balance. Once the reaction starts, the balance shows a negative mass. The mass goes down because the carbon dioxide escapes through the cotton wool.

When we plot a graph of mass of carbon dioxide lost against time, we obtain something similar to the one in Figure 20.6.

The steeper the slope (gradient) of the line, the faster the reaction. We can see from Figure 20.6 that about 0.47 g of carbon dioxide is produced in the first minute. Only about 0.20 g of extra carbon dioxide is produced in the second minute, the reaction is slowing down.

HINT

You are unlikely to be asked to draw tangents and find rates like this in an exam.



▲ Figure 20.6 The mass of carbon dioxide lost when calcium carbonate reacts with hydrochloric acid.

We can calculate the *average rate* of the reaction during any time interval by using

$$\text{rate} = \frac{\text{mass of } CO_2 \text{ lost}}{\text{time}}$$

For example, the average rate of the reaction in the *first* minute

$$= \frac{0.47}{1} = 0.47\,\text{g/min}$$

The average rate of the reaction in the *second* minute

$$= \frac{0.20}{1} = 0.20\,\text{g/min}$$

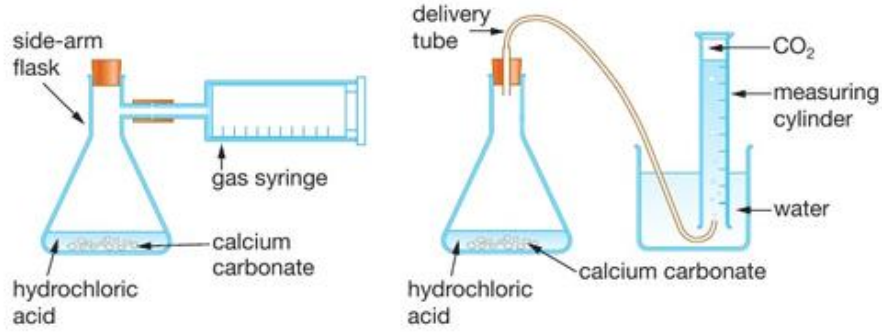
The average rate of the reaction over the *first two* minutes

$$= \frac{0.67}{2} = 0.34\,\text{g/min}$$

We can see the reaction is fastest at the beginning. It then slows down, until it eventually stops because *all the hydrochloric acid has been used up*.

We can measure how fast the reaction is going *at any time point* by finding the slope (gradient) of the line at that point. This is the rate of the reaction *at that point* (rather than the average). This is done by drawing a tangent to the line at the time you are interested in and finding its slope (gradient). For example, at 5 minutes the carbon dioxide is being lost at the rate of about 0.05 g per minute (see Figure 20.6).

We can also follow the rate of this reaction by measuring the *volume* of carbon dioxide given off. The apparatus shown in Figure 20.7 can be used.



▲ Figure 20.7 Either method can be used to measure the volume of CO_2 given off.

REMINDER

Remember, the marble chips were in excess (there was more than enough to react with all the hydrochloric acid). There will still be unreacted marble chips in the flask at the end.

EXPLAINING WHAT'S HAPPENING

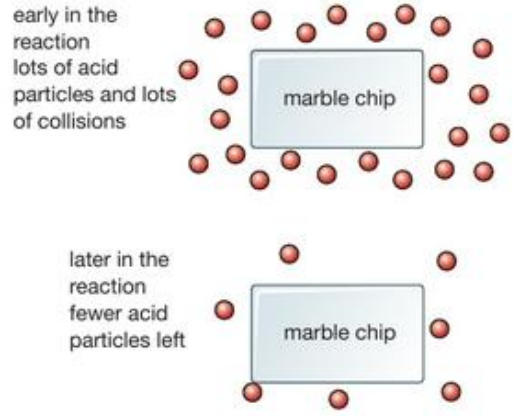
We can explain the shape of the curve by thinking about the particles present and how they interact. This is called the **collision theory**.

Reactions can happen only when particles collide. Not all collisions end up in a reaction. Many particles just bounce off each other. In order for a reaction to actually happen, *the particles have to collide with a minimum amount of energy*, called the **activation energy**. *The collisions with energy greater than or equal to the activation energy* are usually called **successful collisions**.

In the reaction between calcium carbonate and hydrochloric acid, particles in the acid have to collide with the particles at the surface of the marble chips. As the acid particles are getting used up, the collision rate decreases, and so the reaction slows down. The marble is in a large excess so that its shape doesn't change very much during the reaction.

KEY POINT

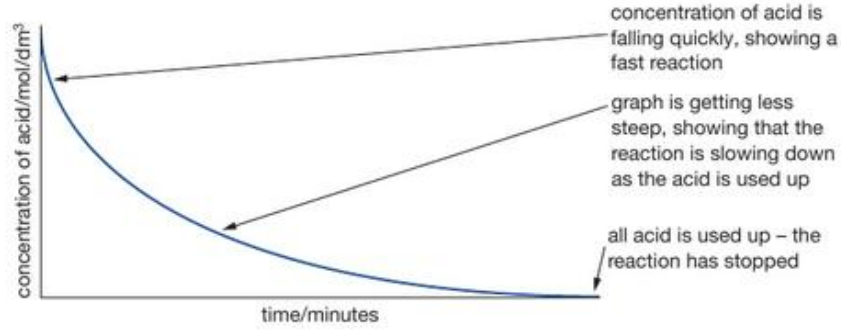
If you have studied Chapter 16, you may realise that the particles we are talking about in the acid are H^+ ions. A reaction occurs when the H^+ ions in the acid collide successfully with the CO_3^{2-} ions in the calcium carbonate. You should not talk about HCl molecules colliding with $CaCO_3$ molecules. If in doubt about what particles are actually colliding, it is probably safer to just use the word *particle*.



▲ Figure 20.8 As the reaction proceeds, the collision rate of acid particles with the marble chips decreases.

A DIFFERENT FORM OF GRAPH

At International GCSE you normally plot graphs showing the mass or volume of product *formed* during a reaction. It is possible, however, that you will see graphs showing the fall in the concentration of one of the *reactants* – in this case, the concentration of the dilute hydrochloric acid.



▲ Figure 20.9 The fall in concentration of hydrochloric acid over time

Where the graph is falling most quickly (is steepest), it shows that the reaction is fastest.

Eventually, the graph becomes horizontal because the reaction has stopped when all the acid has been consumed.

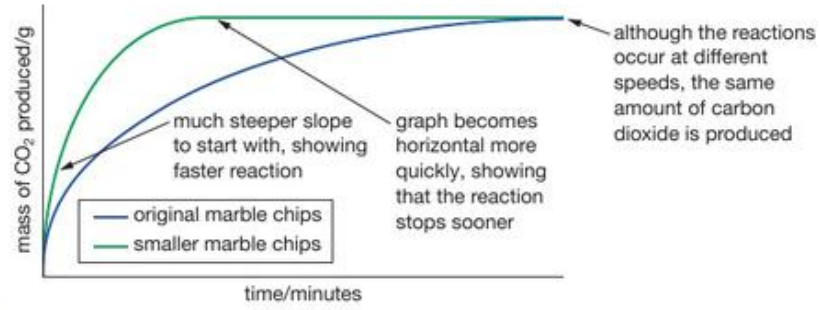
CHANGING THE SURFACE AREA OF THE REACTANTS

ACTIVITY 6

▼ PRACTICAL: INVESTIGATING THE EFFECT OF CHANGING THE SURFACE AREA OF MARBLE CHIPS ON THE RATE OF REACTION BETWEEN MARBLE CHIPS AND DILUTE HYDROCHLORIC ACID

We can repeat the experiment in Activity 5 using exactly the same quantities of everything, but using much smaller marble chips. The reaction with the smaller chips happens faster.

We can plot both sets of results (Activities 5 and 6) on the same graph (Figure 20.10). Notice that the same mass of carbon dioxide is produced because we are using the same quantities of everything in both experiments. However, the reaction with the smaller chips starts off much faster and finishes sooner.



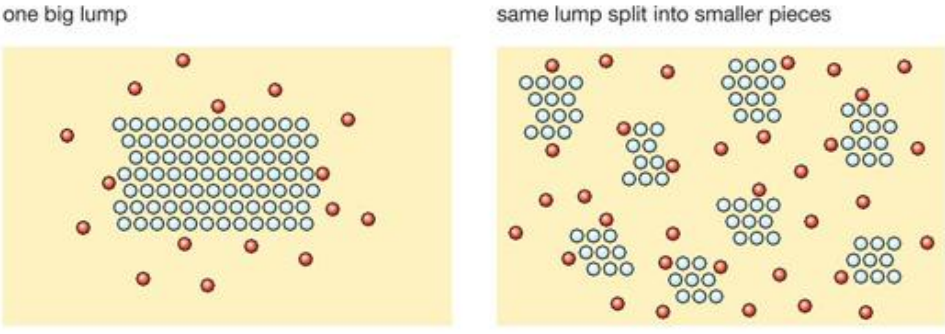
▲ Figure 20.10 The effect of using smaller marble chips

! Safety Note: Wear eye protection and avoid skin contact with the acid.

HINT

If we are going to investigate the effect of changing the size of the marble chips, it is important that everything else stays exactly the same to make this a valid (fair) test.

Reactions between solids and liquids (or solids and gases) are faster if the solids are present as a lot of small pieces rather than a few big ones. The more finely divided the solid, the faster the reaction. This is because the *surface area* in contact with the gas or liquid is *much greater* and there are more particles of the solid exposed on the surface. Only the particles on the surface are available for collisions. The *frequency of successful collisions increases* as the surface area of the solid increases.



▲ Figure 20.11 The more divided the solid, the faster the reaction



▲ Figure 20.12 A catalytic converter has a honeycomb structure to give a very large surface area for the exhaust gases to flow through.

Large surface areas are frequently used to speed up reactions outside the lab. For example, a **catalytic converter** for a car uses expensive metals such as platinum, palladium and rhodium coated onto a honeycomb structure in a very thin layer to give the maximum possible surface area.

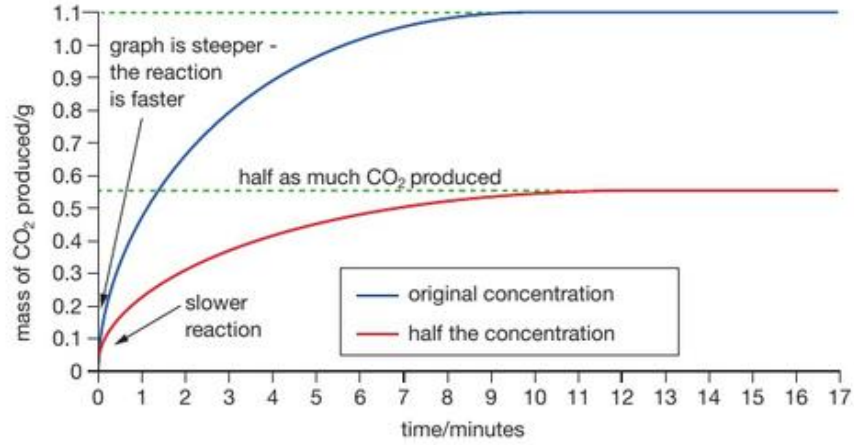
In the presence of these metals, harmful substances such as carbon monoxide and nitrogen oxides are converted into relatively harmless carbon dioxide and nitrogen. The large surface area means the reaction is very rapid. This is important because the gases in the exhaust system are in contact with the catalytic converter for only a very short time.

CHANGING THE CONCENTRATION OF THE REACTANTS

THE EFFECT OF CHANGING THE CONCENTRATION

We can repeat our original experiment with large marble chips and hydrochloric acid. Everything is kept the same except we use hydrochloric acid of half the concentration.

We find that reducing the concentration of the acid makes the reaction slower. We can see this on our graph because the graph (red line) is less steep than for our original experiment (blue line).



▲ Figure 20.13 The effect of changing the concentration of the acid

REMINDER

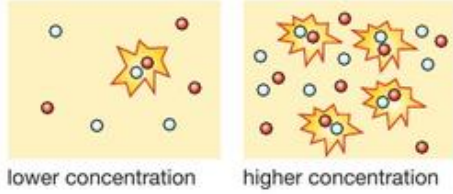
Remember, the marble chips were in excess, so this makes no difference.

In our original experiment we used 25 cm^3 of 2.00 mol/dm^3 hydrochloric acid. In this experiment we used 25 cm^3 of 1.00 mol/dm^3 hydrochloric acid. In the second experiment we have started with half the number of hydrochloric acid particles (half as many moles) and so we will produce half as much carbon dioxide.

KEY POINT

When explaining this, do not only say that there are more collisions. You must always include the idea of time. So, you should say things such as '*the particles collide more frequently or there are more collisions per second*'.

In general, if you increase the concentration of the reactants, the reaction becomes faster. *Increasing the concentration increases the number of acid particles within a fixed volume*, therefore the particles are closer together and collide more frequently. There are *more successful collisions* between the acid particles and the marble chips *every second*.



▲ Figure 20.14 At a higher concentration, more particles collide per second

Safety Note: Wear eye protection and avoid skin contact with the acid.

KEY POINT

To calculate the concentration of hydrochloric acid you need to look at the proportions of acid and water. The equation to calculate the concentration of hydrochloric acid is:

$$\text{concentration (mol/dm}^3\text{)} = \frac{\text{volume of hydrochloric acid (cm}^3\text{)}}{\text{total volume of solution (25 cm}^3\text{)}} \times \text{original concentration (2.00 mol/dm}^3\text{)}$$

DID YOU KNOW?

The words *proportional* and *directly proportional* mean the same thing. If the graph we plotted was a straight line that did not go through the origin, this is not a proportional relationship. In that case we would describe the relationship as *linear*.

EXTENSION WORK

The relationship between the rate of a reaction and the concentration of the reactants is not always as uncomplicated as it seems. In some reactions, the rate of the reaction is proportional to the concentration of the reactants, as shown by the example in Figure 20.15. We say the reaction is first order with respect to a particular reactant.

If you go on to do chemistry at a higher level, you will come across reactions in which increasing the concentration of one of the reactants has no effect on how fast the reaction happens. These reactions are of zero order with respect to the reactants. In some other cases the rate of a reaction might be proportional to the square of the concentration of a reactant. In this case we say the reaction is of second order with respect to the reactant.

ACTIVITY 7

▼ PRACTICAL: INVESTIGATING THE EFFECT OF CHANGING THE CONCENTRATION OF THE ACID ON THE RATE OF REACTION BETWEEN MARBLE CHIPS AND DILUTE HYDROCHLORIC ACID

We can repeat our previous experiment in Activity 5 using the original large marble chips, but using hydrochloric acid of different concentrations. Everything else would be the same, that is, the mass of the marble chips (5.00 g) and the total volume of the acid (25 cm^3). We can dilute the acid by adding distilled water to the original 2.00 mol/dm^3 solution but making sure the total volume of water and acid remains at 25 cm^3.

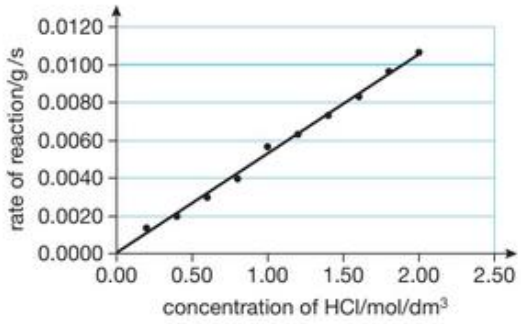
For example, if 12.5 cm^3 of the original HCl is mixed with 12.5 cm^3 of distilled water, we get an acid solution of half of the original concentration (1.00 mol/dm^3). If only 5 cm^3 of the original HCl is mixed with 20 cm^3 of distilled water, the acid solution is now one-fifth (5 out of 25) of the original concentration (0.40 mol/dm^3).

In this experiment we are going to calculate the average rate during the first 30 seconds, so we record the mass of carbon dioxide lost in 30 seconds.

We can calculate the average rate of reaction within the first 30 seconds by dividing the mass loss by 30. For example, if we obtain a loss of 0.32 g:

$$\text{average rate} = \frac{0.32}{30} = 0.011\ \text{g/s}$$

To identify the effect of changing concentration on rate, we can plot the results for the different concentrations on a piece of graph paper with rate on the *y*-axis and concentration of acid on the *x*-axis (Figure 20.15). The line of best fit should go through the origin (0, 0). If there is no acid, there should be no rate. A straight line going through the origin shows that the rate of the reaction is *directly proportional* to the concentration of the hydrochloric acid. If the concentration doubles, the rate should double. If the concentration triples, the rate should triple.

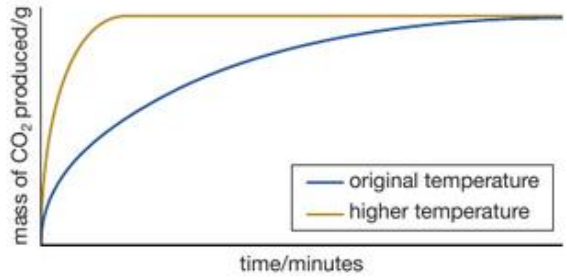


▲ Figure 20.15 The more concentrated the acid, the faster the reaction

CHANGING THE TEMPERATURE OF THE REACTION

We can do the original experiment again, but this time at a higher temperature. We keep everything else exactly the same as before.

Reactions get faster as the temperature is increased; the graph is *steeper* and *finishes sooner*. The same mass of gas is given off because we have used the same quantities of everything in the mixture.



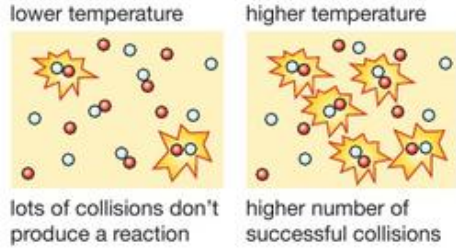
▲ Figure 20.16 The effect of changing the temperature of the reaction

There are two factors that need to be considered when explaining why increasing temperature increases the rate of reaction:

- Increasing the temperature means that the particles are moving faster, and so collide more frequently. That will make the reaction go faster, but it only accounts for a small part of the increase in rate.
- We learned above that in order for a collision to cause a reaction (*a successful collision*), the particles have to collide with a minimum amount of energy, called the *activation energy*. A relatively small increase in temperature produces *a very large increase in the number of particles with energy greater than or equal to the activation energy*. This means that it is going to be much more probable that two particles which have sufficient energy to react collide with each other. So *the frequency of successful collisions increases*.

KEY POINT

As an approximation, a 10 °C increase in temperature approximately doubles the rate of a reaction. This works for quite a lot of reactions that we meet in the lab, as long as the temperature remains within a few tens of degrees of room temperature.



▲ Figure 20.17 A small increase in temperature produces a large increase in the number of collisions with energy greater than the activation energy.

CHANGING THE PRESSURE ON THE REACTION

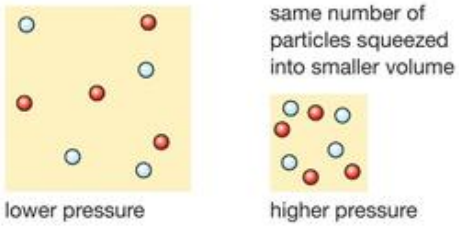
KEY POINT

In solids and liquids, particles are packed closely together and there is hardly any space in between. Increasing the pressure would not change the volumes of the solids or liquids much, therefore there would hardly be any change in the number of particles in a certain volume.

Changing the pressure of a reaction in which the reactants are only solids or liquids makes almost no difference to the rate of reaction. But increasing the pressure of a reaction where the reactants are *gases* does speed up the reaction.

If we have a fixed mass of a gas, we increase the pressure by squeezing it into a smaller volume.

This forces the particles *closer together*, so they *collide more frequently*. This is exactly the same as increasing the concentration of the gas.



▲ Figure 20.18 Increased pressure means gas particles collide more frequently.

CATALYSTS

WHAT ARE CATALYSTS?

Catalysts are *substances that speed up chemical reactions*, but aren't used up in the process. They are still there, *chemically unchanged, at the end of the reaction*. Because they aren't consumed, small amounts of catalyst can be used to process lots and lots of reactant particles. Different reactions need different catalysts.

THE CATALYTIC DECOMPOSITION OF HYDROGEN PEROXIDE



▲ Figure 20.19 Bombardier beetles use hydrogen peroxide as part of their defence mechanism.

Bombardier beetles defend themselves by spraying a hot, unpleasant liquid at their attackers. Part of the reaction involves splitting hydrogen peroxide into water and oxygen, using the enzyme catalase. **Enzymes** are *biological catalysts*. This reaction happens almost explosively, and produces a lot of heat. The same enzyme can be found in potatoes, or even liver tissues.

There are a lot of other things that also catalyse the decomposition of hydrogen peroxide. Some examples are manganese(IV) oxide (also called manganese dioxide), MnO_2, and lead(IV) oxide, PbO_2. Manganese(IV) oxide is what is normally used in the lab to speed up the decomposition of hydrogen peroxide.

The reaction happening with the hydrogen peroxide is:

hydrogen peroxide → water + oxygen

$$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$$

Notice that we don't write catalysts into the equation because they are chemically unchanged at the end of the reaction. If you like, you can write their name or formula over the top of the arrow.

SHOWING THAT A SUBSTANCE IS A CATALYST



▶ Figure 20.20 The beakers both contain hydrogen peroxide solution, the right-hand one has MnO_2 added to speed up oxygen production.

It isn't difficult to show that manganese(IV) oxide speeds up the decomposition of hydrogen peroxide to produce oxygen. Figure 20.20 shows two beakers, both of which contain hydrogen peroxide solution. Without the catalyst, there is only a trace of bubbles in the solution. With it, oxygen is given off quickly.

How can you show that the manganese(IV) oxide is chemically unchanged by the reaction? It still looks the same, but has any been used up? You can only find out by weighing it before you add it to the hydrogen peroxide solution and then reweighing it at the end.

You can separate the manganese(IV) oxide from the liquid by filtering it through a weighed filter paper, allowing the paper and residue to dry, and then reweighing to calculate the mass of the remaining manganese(IV) oxide. You should find that the mass hasn't changed.

Safety Note: Wear eye protection and avoid skin contact with the solution.

HINT

The experiment in Figure 20.21 is very simple and easy to set up. However, using this setup has the disadvantage that some oxygen will escape at the beginning of the reaction when MnO_2 is added to H_2O_2 before the bung can be put back on the conical flask. Using a weighing bottle like the one in Figure 20.22 is a simple way of mixing the chemicals together without losing any oxygen before you can get the bung in. When you are ready to start the reaction, shake the flask so that the weighing bottle falls over and the manganese(IV) oxide comes into contact with the hydrogen peroxide. You need to keep shaking so that an even mixture is formed.

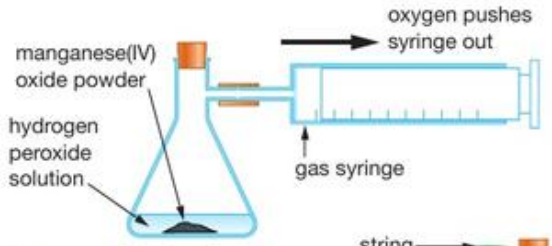
DID YOU KNOW?

'*vol*', meaning volume, is a measurement for concentration for hydrogen peroxide. 1 cm^3 of 2 vol hydrogen peroxide solution decomposes completely to give 2 cm^3 of oxygen.

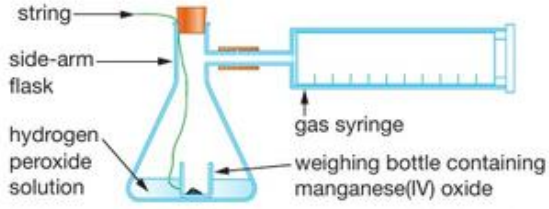
ACTIVITY 8

▼ PRACTICAL: INVESTIGATE THE EFFECT OF DIFFERENT SOLIDS ON THE CATALYTIC DECOMPOSITION OF HYDROGEN PEROXIDE SOLUTION

Figure 20.21 shows apparatus that can be used to measure how the volume of oxygen produced changes with time.



▲ Figure 20.21 Apparatus to measure the volume of oxygen evolved.

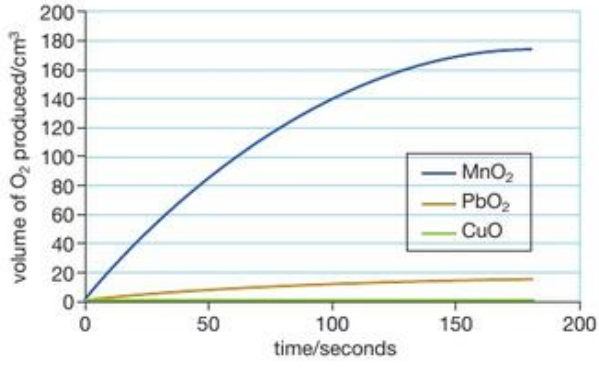


▲ Figure 20.22 Modified apparatus to avoid the loss of oxygen at the beginning of the experiment.

The following procedure could be used:

- Measure 100 cm^3 of 2 *vol* hydrogen peroxide and transfer to a 250 cm^3 conical flask.
- Weigh out 0.20 g of manganese(IV) oxide on a balance.
- Add the manganese(IV) oxide to the hydrogen peroxide and quickly replace the bung with the gas syringe already attached. Swirl the reaction mixture at a constant speed.
- Record the amount of oxygen produced every 20 seconds for 3 minutes and plot a graph of volume of oxygen versus time.
- Repeat the reaction with 0.20 g of lead(IV) oxide and copper(II) oxide but keep everything else the same.

The sample data below show manganese(IV) oxide is a very effective catalyst for the decomposition of hydrogen peroxide. In comparison, lead(IV) oxide is less effective as the rate of the reaction is much slower and copper(II) oxide does not act as a catalyst for this reaction at all.



▲ Figure 20.23 The catalytic effect of different solids on the decomposition of H_2O_2.